

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2145

Roanoke Rapids 345 kV
March 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2145. The Interconnection Customer proposes a 284.9 MW solar plus storage facility interconnecting at the Roanoke Rapids 345 kV substation in the North Carolina area. The request was received on July 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 571 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

This project does not cause any thermal violations under system intact or single contingency conditions. Estimated network upgrade cost: \$0

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

No overloads identified.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.